Forest Harvesting

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Forest Harvesting (A08962)

Short Title:	Forest Harvesting		
Faculty	Science and Computing		
Department	Science		
Cluster	Forestry		
Author	TKENT		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

Sustainable production of timber, fibre and fuel from forests is dependent on implementing the appropriate harvesting methods and systems on each site. Harvesting operations impact forest regeneration, forest health and productivity over time. Harvesting must be implemented ensuring safety of staff and the public and protection of soil, water, biodiversity and archaeological features, in accordance with regulatory guidelines. Forest harvesting in Ireland is forecast to expand from the current three million cubic metres per annum to six million cubic metres per annum over the next fifteen years, according to the All Ireland Roundwood Production Forecast 2011 to 2028, published by COFORD. Expanding professional capacity in this area will be vital in realising this expansion opportunity. In this module students will gain specialist knowledge on the harvesting methods and systems employed in sustainable harvesting, and the limitations of these systems due to site conditions. Students will gain skills in designing a harvest plan, assessing harvesting productivity and production cost, carrying out an Environmental Impact Assessment, diagnosing site specific safety issues and constraints on harvesting, and identifying appropriate solutions. To enable students to carry out this work, they will be given an in-depth knowledge of harvesting-related equipment, safety procedures, machine productivity and operation cost and environmental issues.

Programmes

`FORS-0012`	BSc in Forestry (`WD_SFORS_D`)	3 / 5 / M

Indicative Content

- Silvicultural systems and regeneration harvesting: clearfell, shelterwood, single tree and group selection, continuous cover
- Harvesting methods: cut to length, wholetree, pole length
- Harvesting systems and equipment for the harvesting, processing and extraction of timber products
- Harvester and forwarder productivity and cost in cut to length thinnings and clearfells
- Machine/ground interaction, ground pressure and traction and machine impact on remaining standing trees
- Environmental considerations and guidelines
- Harvest planning and scheduling: Harvest Plan & Environmental Impact Assessment development
- Thinning: thinning types, thinning control, thinning intensity, timing and impact;
- Machine cost calculations: ownership and operating costs, utilisation rate and hourly machine cost
- Assessing machine productivity: time and motion studies, assessing production over time, determining unit cost of timber production
- Managing operator and public health and safety on a harvesting site
- Information technology in harvesting and timber haulage, harvest head calibration, assortment specifications & the Irish timber market

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Differentiate between different harvesting methods and systems and the key components of harvesters and forwarders.
2. Apply the appropriate harvesting methods and systems under various harvest site conditions taking into consideration the principles of SFM.
3. Demonstrate knowledge of the component costs of machine ownership and operation, and calculate the hourly machine rate from machine cost and utilisation rate.
4. Appraise the key points in harvester head calibration and the information stream from pre-sale inventory to timber haulage.
5. Recommend the correct working procedures on a site in accordance with the Forest Service Environmental Guidelines.
6. Construct a harvesting plan for a clearfell/thinning area taking into consideration operator safety, operational aspects, and the protection of the environment.
7. Prepare a technical report on the productivity and timber production cost of a harvester and forwarder operation.
8. Compile a risk assessment containing the appropriate precautionary measures in order to manage operator and public health and safety on a harvesting site.

Learning and Teaching Methods

- Lectures: will be used to introduce the principles and practices used in forest harvesting and the regulatory framework around the management of forest operations with regard to sustainable forest management, environmental protection and health and safety.
- Field trips: will provide students with opportunities for experiential learning of forest harvesting operations management and health and safety; students will also gather case study material and data sets for the harvesting plan and productivity study project.
- Independent Learning: Reading recommended reference books, supplementary papers and handouts, and viewing recommended web-based material to consolidate knowledge of the module content.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		1, 2, 3, 4, 5
Continuous Assessment	50%		
Project		20%	6,8
Project		20%	7
Case Studies		10%	8

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- Anon., A. _Code of Best Forest Practice - Ireland_. Co. Wexford, Ireland: Forest Service, 2000.
- Anon., A. _Code of Practice for Managing Safety and Health in Forestry Operations_. Dublin, Ireland: Health and Safety Authority, 2009.
- Anon., A. _Forest Harvesting and the Environment Guidelines_. Co. Wexford, Ireland: Forest Service, 2000.
- Magagnotti, N. and R. Spinelli . _Good practice guidelines for biomass production studies_ . Sesto Fiorentino (FI) ITALY: CNR IVALSA, 2012.

Supplementary Material

- "InnovaWood, Dublin." Harvesthead. http://77.74.50.56/Innovawood/DesktopDefault.aspx?tabindex=15&tabid=162
- "Supplementary reading of selected research papers and forest operation management reports will be made available to students on Moodle: Forest Harvesting module page." WIT Moodle. www.wit.ie/moodle
- Owende, P.M.O., J. Lyons and S.M. Ward. _Operations Protocol for ECO-Efficient Wood Harvesting on Sensitive Sites_. UCD Dublin Ireland: Dept. of Eng., UCD, 2002.

Requested Resources

- Lecture Room: Loose Seated